

CyberQ Shield (CNAPP Features)

*An Internship Project Report submitted in partial fulfilment of the requirements
for the award of the degree of*

Master of Computer Applications

By

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DECLARATION

I hereby declare that the work which is being presented in the M.C.A. Project “**CyberQ Shield(CNAPP Features)**”, in partial fulfillment of the requirements for the award of the **Master of Computer Applications** and submitted to the Department of Computer Engineering and Applications of GLA University, Mathura, is an authentic record of my own work carried under the supervision of **Mr. Harshit Bisht**.

The contents of this project report, in full or in parts, have not been submitted to any other Institute or University for the award of any degree.

Sign _____

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CERTIFICATE

This is to certify that the above statements made by the candidate are correct to the best of my/our knowledge and belief.

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ABSTRACT

The proliferation of cloud-native applications has fundamentally transformed modern software deployment, bringing new security and operational challenges that traditional DevOps pipelines inadequately address. This internship project, undertaken at Seclogic Technology Private Limited, focuses on the implementation of CyberQ Shield — a Cloud-Native Application Protection Platform (CNAPP) — encompassing CI/CD pipeline automation, Kubernetes orchestration, infrastructure as code, container security, and end-to-end observability.

The project established a fully automated CI/CD pipeline using Jenkins and GitLab CI/CD, integrating build, test, Docker image creation, Trivy vulnerability scanning, and zero-downtime Kubernetes deployment. AWS infrastructure was provisioned via Terraform, while the ELK Stack (Elasticsearch, Logstash, Kibana) and Prometheus/Grafana provided centralised logging and real-time monitoring. The platform embeds security at every lifecycle stage — operationalising "shift-left" security through mandatory pre-deployment container scanning.

All twenty test cases across CI/CD, containerisation, Kubernetes, security scanning, and observability achieved a 100% pass rate. The project demonstrates that integrating security, automation, and observability within a unified cloud-native platform delivers measurable improvements in deployment speed, reliability, and security posture.

Keywords: *CNAPP, DevOps, DevSecOps, CI/CD, Kubernetes, Docker, Terraform, AWS, Jenkins, GitLab, ELK Stack, Prometheus, Grafana, Trivy, Infrastructure as Code, Container Security.*

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Chapter 1

Introduction

1.1 Introduction to DevOps

DevOps, an amalgamation of "Development" and "Operations," represents a cultural and technical paradigm that aims to shorten the software development lifecycle while delivering features reliably and repeatedly. By dismantling traditional organisational silos between development and operations teams, DevOps fosters shared ownership of the entire delivery pipeline — from code commit to production deployment.

Key pillars of DevOps include Continuous Integration (CI), Continuous Delivery (CD), Infrastructure as Code (IaC), monitoring, and observability. The modern DevOps toolchain employed in this project spans GitLab (version control), Jenkins (pipeline orchestration), Docker (containerisation), Kubernetes (orchestration), Terraform (IaC), AWS (cloud platform), Prometheus/Grafana (monitoring), and the ELK Stack (logging).

1.2 Introduction to CNAPP

A Cloud-Native Application Protection Platform (CNAPP) is a holistic security framework that consolidates multiple cloud security capabilities — Cloud Security Posture Management (CSPM), Cloud Workload Protection Platform (CWPP), Infrastructure Entitlement Management (CIEM), and Kubernetes Security Posture Management (KSPM) — into a single integrated solution. CNAPP enables comprehensive visibility, misconfiguration detection, vulnerability management, and compliance enforcement across the entire cloud-native application lifecycle.

CyberQ Shield represents a practical CNAPP implementation within a real-world DevSecOps pipeline, demonstrating how security can be seamlessly integrated at every stage of the software delivery process.

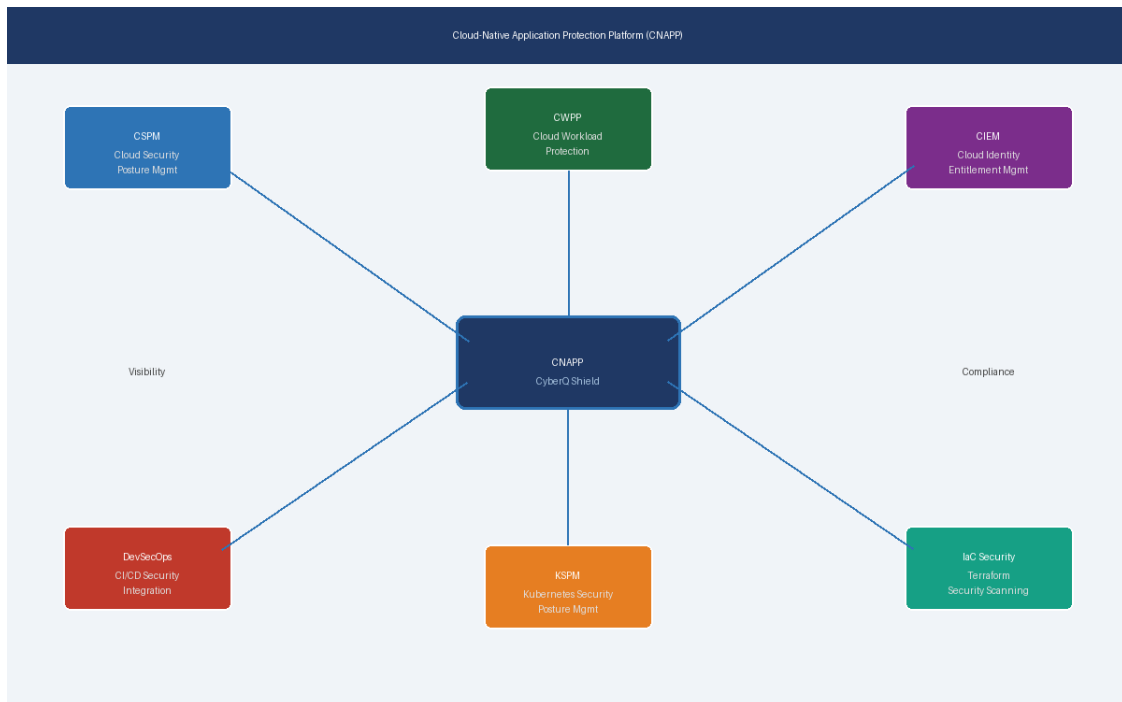


Fig. 1.1: CNAPP Architecture – Components Overview (CyberQ Shield)

1.3 Problem Statement

The rapid adoption of cloud-native infrastructure has created security and operational challenges that traditional pipelines inadequately address:

- Absence of automated pre-deployment security scanning, resulting in vulnerable container images reaching production.
- Fragmented monitoring and logging tooling leading to delayed incident detection and poor root-cause analysis.
- Manual and error-prone infrastructure provisioning increasing the risk of misconfigurations.
- Inability to enforce security policies consistently across multi-stage pipelines, leading to compliance gaps.

The CyberQ Shield project was conceived to address these challenges by constructing an end-to-end, automated, security-integrated DevOps pipeline.

1.4 Objectives of the Project

1. Design and implement a fully automated CI/CD pipeline integrating build, test, security scan, and zero-downtime deployment.
2. Provision AWS cloud infrastructure using Terraform for reproducible, version-controlled IaC management.
3. Deploy containerised applications on Kubernetes with appropriate resource management and scaling.
4. Integrate Trivy vulnerability scanning to detect and block insecure container images before deployment.
5. Establish real-time monitoring and alerting using Prometheus and Grafana.
6. Implement centralised log aggregation and analysis using the ELK Stack.
7. Demonstrate practical CNAPP principles in a production-grade cloud-native environment.

1.5 Scope and Report Organisation

The scope encompasses: Infrastructure (AWS/Terraform), Containerisation (Docker), Orchestration (Kubernetes), CI/CD (Jenkins/GitLab/ArgoCD), Security (Trivy/RBAC), and Observability (Prometheus/Grafana/ELK). The report is organised in seven chapters: Chapter 2 — requirements; Chapter 3 — system design; Chapter 4 — implementation; Chapter 5 — testing; Chapters 6 and 7 — conclusion and future scope.

1.6 Summary of Similar Systems

Several commercial and open-source platforms offer partial CNAPP capabilities. Prisma Cloud provides comprehensive CSPM and CWPP but requires significant licensing investment. Aqua Security focuses on container security with runtime protection. Sysdig Secure offers eBPF-powered Kubernetes runtime security. Snyk provides developer-first container scanning but lacks full CNAPP coverage. CyberQ Shield differentiates itself by delivering an integrated, open-source-oriented solution unifying CI/CD, container security, IaC, and full-stack observability in a practitioner-focused implementation.

Chapter 2

Software Requirement Analysis

2.1 Functional Requirements

The system is designed to support a comprehensive DevOps workflow by integrating automation, security, and monitoring capabilities across the software delivery lifecycle. It shall automate the build, testing, and deployment processes through a CI/CD pipeline while provisioning AWS infrastructure using Terraform for consistency and scalability. The system will deploy containerised applications on Kubernetes clusters, ensuring efficient orchestration and management. Security is enforced by scanning container images for vulnerabilities and implementing Kubernetes RBAC along with secure secret management. Additionally, the system shall provide real-time monitoring by collecting and visualising metrics, as well as aggregating and analysing logs from all services.

Req. ID	Requirement	Priority	Module
FR-01	Automate build, test, and deployment via CI/CD pipeline	High	CI/CD
FR-02	Scan container images for vulnerabilities before deployment	High	Security
FR-03	Provision AWS infrastructure automatically using Terraform	High	Infrastructure
FR-04	Deploy containerised applications on Kubernetes clusters	High	Kubernetes
FR-05	Collect and visualise metrics in real time	High	Monitoring
FR-06	Aggregate, index, and analyse logs from all services	Medium	Logging
FR-07	Trigger automated alerts on threshold breaches	High	Alerting
FR-08	Enforce Kubernetes RBAC and secret management	High	Security
FR-09	Support horizontal pod auto-scaling	Medium	Kubernetes
FR-10	Provide pipeline audit trail and execution status	Medium	CI/CD
FR-11	Support multi-stage Docker builds for optimised images	Medium	Containerisation
FR-12	Enable Kubernetes deployment rollback on failure	High	Kubernetes

Table 2.1: Functional Requirements of CyberQ Shield

2.2 Non-Functional Requirements

The system is designed to meet key non-functional requirements that ensure scalability, availability, performance, security, reliability, maintainability, and observability. It shall support scalability by enabling Kubernetes Horizontal Pod Autoscaling to dynamically adjust up to 10 replicas per deployment based on CPU and memory usage. To ensure availability, core platform services must maintain a minimum uptime of 99.5% as defined by the SLA. The system's performance is optimised to complete the full CI/CD pipeline within 15 minutes end-to-end. From a security perspective, all production container images must be free from critical vulnerabilities, enforcing a zero critical CVE policy through automated gates. Reliability is ensured by implementing liveness and readiness probes across 100% of pods. For maintainability, all infrastructure components are managed as code using Terraform, achieving complete IaC coverage.

Category	Requirement	Target Metric
Scalability	Kubernetes HPA scales pods on CPU/memory load	Up to 10 replicas per deployment
Availability	Core platform services maintain high availability	99.5% uptime SLA
Performance	Full CI/CD pipeline execution time	< 15 minutes end-to-end
Security	No Critical CVEs in production images	Zero Critical CVEs enforced by gate
Reliability	All pods have liveness and readiness probes	100% probe coverage
Maintainability	All infrastructure managed as code	100% IaC coverage via Terraform
Observability	Metric data retained for trend analysis	30-day Prometheus retention
Log Retention	Logs retained for compliance and forensics	90-day Elasticsearch retention

Table 2.2: Non-Functional Requirements

2.3 Feasibility Study

2.3.1 Technical Feasibility

All selected technologies — Docker, Kubernetes, Jenkins, GitLab CI/CD, Terraform, AWS, Prometheus, Grafana, and ELK — are mature, extensively documented platforms. The technical team at Seclogic possesses requisite expertise and the internship provided appropriate cloud infrastructure and tooling access.

2.3.2 Economic, Operational & Time Feasibility

The project leverages primarily open-source technologies, significantly reducing licensing costs. AWS costs are managed through right-sized instances and auto-scaling. The three-month internship provided sufficient time to design, implement, test, and document all core components across iterative development sprints.

Feasibility Dimension	Assessment	Remarks
Technical	Feasible	Mature tools; strong community and documentation support
Economic	Feasible	Open-source stack; AWS costs managed via right-sizing
Operational	Feasible	Team has skills; modular architecture aids maintainability
Time	Feasible	3-month internship sufficient for full implementation and testing

Table 2.3: Feasibility Analysis Summary

2.4 Module Description

- CI/CD Module: Jenkins and GitLab CI/CD pipeline configurations automating the complete build, test, scan, and deployment lifecycle.
- Kubernetes Module: Cluster provisioning, namespace organisation, Deployment manifests, HPA, and health probe configuration.
- Monitoring Module: Prometheus metric collection, Grafana dashboard creation, and Alertmanager notification routing.
- Logging Module: ELK Stack — Filebeat log collection, Logstash transformation, Elasticsearch indexing, and Kibana visualisation.
- Security Module: Trivy container scanning, Kubernetes RBAC, Secrets management, and network policy enforcement.
- Infrastructure Module: Terraform-managed AWS resources — VPC, EC2, security groups, IAM, S3, and ELB.

Chapter 3

System Design

3.1 System Architecture & Technology Stack

The CyberQ Shield architecture is a multi-layered cloud-native platform integrating the software delivery pipeline, container orchestration, observability, and security. The architecture operates across five layers: Developer Layer (GitLab code commits), CI/CD Orchestration Layer (Jenkins/GitLab CI runners), Infrastructure Layer (AWS resources via Terraform), Orchestration Layer (Kubernetes cluster), and Observability Layer (Prometheus, Grafana, ELK Stack).

Category	Technology	Purpose
Cloud Platform	Amazon Web Services (AWS)	EC2, VPC, S3, ELB infrastructure hosting
IaC	Terraform	Declarative infrastructure provisioning
Containerisation	Docker	Application containerisation and image management
Orchestration	Kubernetes	Container scheduling, scaling, and lifecycle
CI/CD	Jenkins + GitLab CI/CD	Pipeline automation and build orchestration
GitOps	ArgoCD	Declarative GitOps continuous deployment
Monitoring	Prometheus + Grafana	Metrics, dashboards, and alerting
Logging	ELK Stack	Log ingestion, indexing, and visualisation
Security Scanning	Trivy (Aqua Security)	Container image vulnerability scanning
OS	Ubuntu 22.04 LTS	Base OS for EC2 instances and containers

Table 3.1: Technology Stack Summary

3.2 CI/CD Workflow Design

The CI/CD workflow initiates when a developer pushes code to a designated GitLab branch. A webhook triggers the CI/CD runner which executes: Checkout → Build & Test → Docker Build → Trivy Scan → Push to Registry → Kubernetes Deploy → Post-deployment Verify. Pipeline failures trigger immediate Slack/email notifications. ArgoCD continuously reconciles the Kubernetes cluster state with the Git repository.

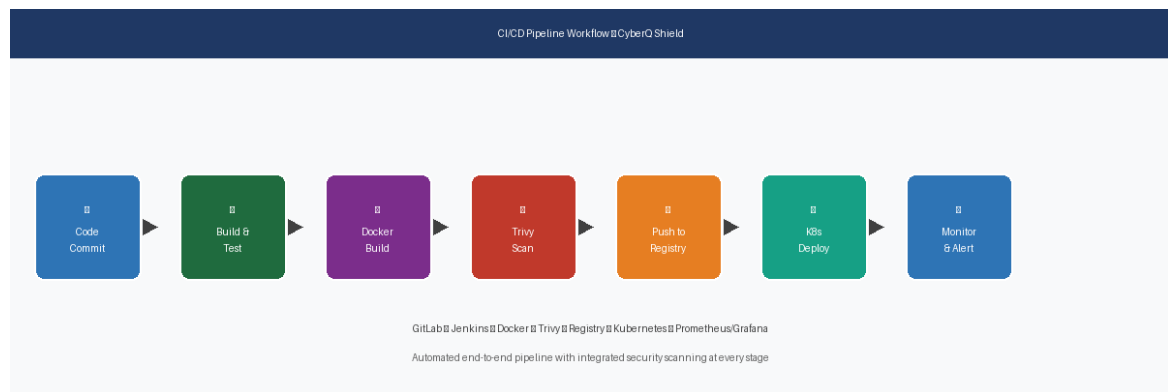


Fig. 3.1: CI/CD Pipeline Workflow Diagram – CyberQ Shield

3.3 Kubernetes Cluster Architecture

The Kubernetes cluster comprises a dedicated Control Plane (API Server, etcd, Scheduler, Controller Manager) and multiple Worker Nodes running kubelet, kube-proxy, and containerd. The cluster is organised into namespaces — production, staging, monitoring, and logging — with resource quotas and RBAC policies. All nodes run on AWS EC2 instances within a private VPC subnet, provisioned by Terraform.

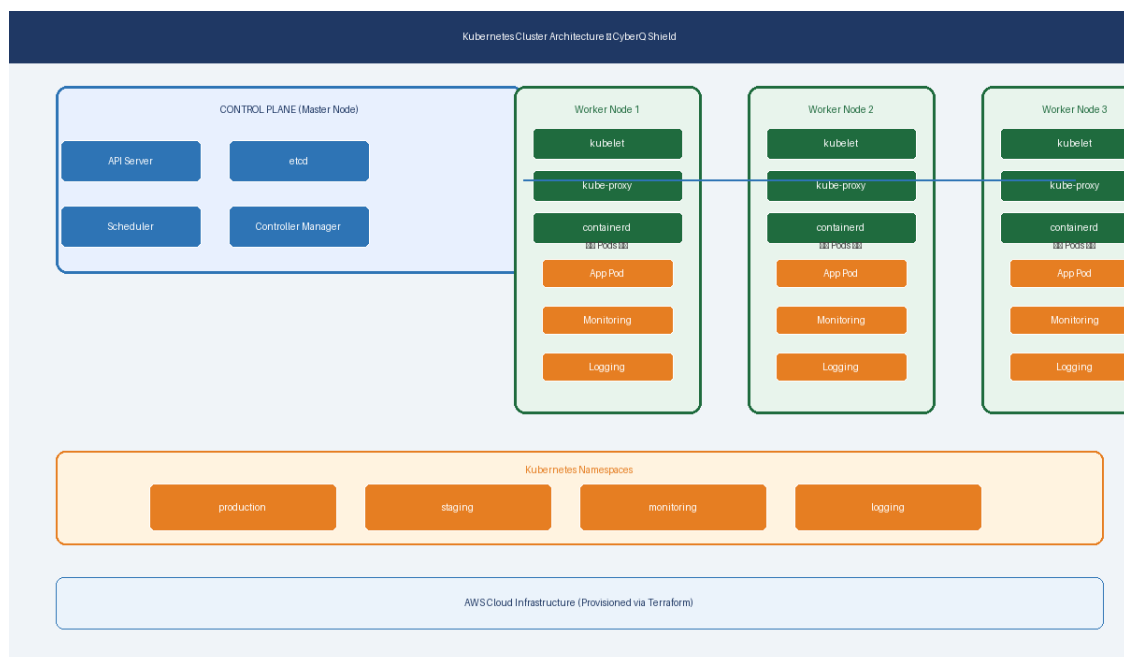


Fig. 3.2: Kubernetes Cluster Architecture Diagram

3.4 Use Case Diagram

The system has four primary actors: Developer (commits code, views pipeline status, accesses Kibana logs), DevOps Engineer (configures pipelines, manages cluster and infrastructure), Security Analyst (reviews Trivy reports, monitors alerts, enforces compliance), and System Administrator (manages RBAC, oversees infrastructure health).

[Fig. 3.3: Use Case Diagram – Insert draw.io / PlantUML diagram here]

Fig. 3.3: Use Case Diagram – CyberQ Shield CNAPP System

3.5 Data Flow Diagrams

3.5.1 DFD Level 0 – Context Diagram

The Level 0 DFD represents CyberQ Shield as a single process receiving code commits from the Developer, infrastructure configurations from the DevOps Engineer, and external CVE database feeds. Outputs include deployment artefacts to Kubernetes, security reports to the Security Analyst, and operational dashboards to all authorised users.

[Fig. 3.4: DFD Level 0 – Context Diagram – Insert diagram here]

Fig. 3.4: DFD Level 0 – Context Diagram

3.5.2 DFD Level 1

The Level 1 DFD decomposes the system into: (1) Code Integration & Build, (2) Security Scanning, (3) Container Image Management, (4) Infrastructure Provisioning, (5) Kubernetes Deployment, and (6) Monitoring & Logging. Data flows represent build artefacts, scan reports, container images, Terraform state files, Kubernetes manifests, and metric/log data.

[Fig. 3.5: DFD Level 1 – Insert diagram here]

Fig. 3.5: DFD Level 1 – System Sub-processes

3.6 Sequence Diagram

The Sequence Diagram traces temporal interactions during deployment: git push → GitLab webhook → CI Runner → Docker Build → Trivy Scan → Registry Push → kubectl apply → Kubernetes API Server → Pod scheduling on Worker Node → Health check validation. Failures at any stage trigger rollback and notifications.

[Fig. 3.6: Sequence Diagram – Deployment Flow – Insert diagram here]

Fig. 3.6: Sequence Diagram – Application Deployment Flow

Chapter 4

Implementation

4.1 Infrastructure Setup on AWS

All AWS resources were provisioned using Terraform with remote state in S3 and DynamoDB state locking for safe team collaboration.

Resource	Configuration	Purpose
EC2 Master Node	t3.medium – 2 vCPU, 4GB, 30GB SSD	Kubernetes Control Plane
EC2 Worker × 2	t3.large – 2 vCPU, 8GB, 50GB SSD	Kubernetes Worker Nodes
EC2 Jenkins Server	t3.medium – 2 vCPU, 4GB, 30GB SSD	Jenkins CI/CD Server
VPC	CIDR: 10.0.0.0/16	Isolated network environment
S3 Bucket	cyberq-terraform-state	Terraform remote state storage
App Load Balancer	HTTP/HTTPS listener	External traffic distribution

Table 4.1: AWS EC2 Instance Configuration

```
# main.tf - Kubernetes Master Node Provisioning
resource "aws_instance" "k8s_master" {
  ami           = var.ubuntu_ami_id
  instance_type = "t3.medium"
  subnet_id     = aws_subnet.private_subnet.id
  key_name      = var.key_pair_name
  root_block_device { volume_size = 30; volume_type = "gp3" }
  tags = { Name = "CyberQ-K8s-Master"; Environment = "production" }
}
```

4.2 CI/CD Pipeline – Jenkins & GitLab

```
// Jenkinsfile - CyberQ Shield
pipeline {
  agent { label "docker-agent" }
  environment { IMAGE_TAG = "${env.BUILD_NUMBER}"; REGISTRY =
"registry.gitlab.com/seclogic/cyberq" }
  stages {
    stage("Checkout") { steps { checkout scm } }
    stage("Build & Test") { steps { sh "mvn clean package"; junit
"target/surefire-reports/*.xml" } }
    stage("Docker Build") { steps { sh "docker build -t
${REGISTRY}:${IMAGE_TAG} ." } }
    stage("Trivy Scan") {
      steps { sh "trivy image --severity CRITICAL,HIGH --exit-code 1 --format
json -o trivy-report.json ${REGISTRY}:${IMAGE_TAG}"
    }
  }
}
```

```

        archiveArtifacts "trivy-report.json" }
    }
    stage("Push") { steps { sh "docker push ${REGISTRY}:${IMAGE_TAG}" } }
    stage("Deploy") { steps { sh "kubectl set image deployment/cyberq-app
app=${REGISTRY}:${IMAGE_TAG} --record" } }
    }
    post {
        failure { slackSend channel:"#devops-alerts", message:"FAILED:
${env.JOB_NAME}" }
        success { slackSend channel:"#devops-alerts", message:"SUCCESS:
${env.JOB_NAME}" }
    }
}

```

[Fig. 4.1: Jenkins Pipeline Stage View – Insert Screenshot Here]

Fig. 4.1: Jenkins Pipeline Execution – Stage View

4.3 ArgoCD – GitOps Deployment

ArgoCD was deployed on the Kubernetes cluster to implement GitOps-based continuous delivery. ArgoCD continuously monitors the Git repository and automatically reconciles the cluster state with the declared desired state, enabling automated drift detection and Git-based rollback. The screenshot below shows the ArgoCD interface accessed during the internship.

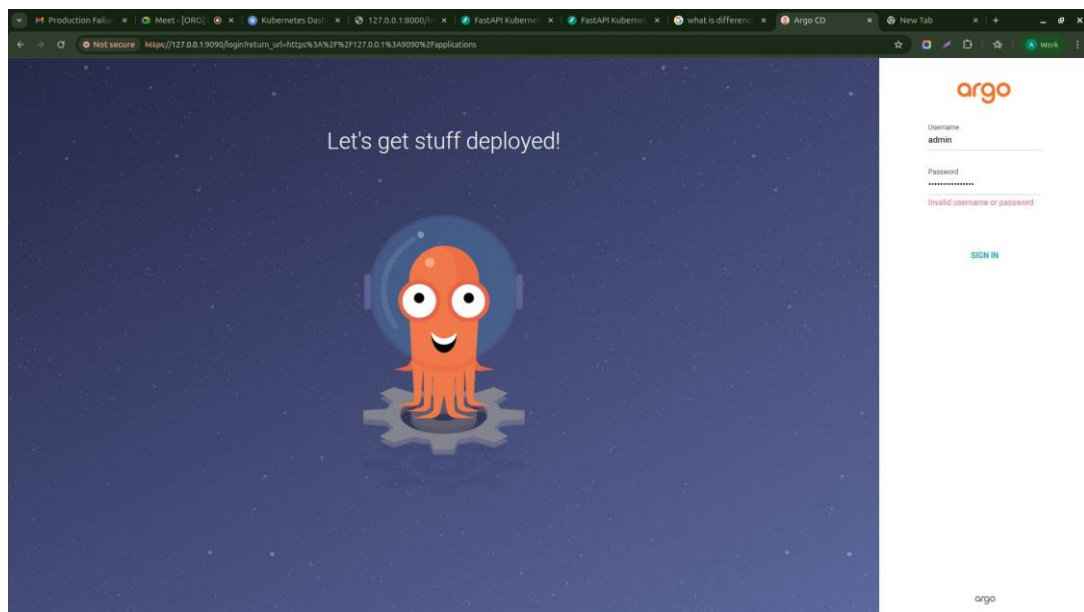


Fig. 4.2: ArgoCD Login Interface – GitOps Continuous Deployment Tool

4.4 Containerisation with Docker

```
# Dockerfile - Multi-Stage Build
FROM maven:3.9-eclipse-temurin-17-alpine AS builder
WORKDIR /app
COPY pom.xml . && RUN mvn dependency:go-offline
COPY src ./src && RUN mvn clean package -DskipTests

FROM eclipse-temurin:17-jre-alpine AS runtime
RUN addgroup -S appgroup && adduser -S appuser -G appgroup
WORKDIR /app
COPY --from=builder /app/target/cyberq-shield-*.jar app.jar
USER appuser
EXPOSE 8080
HEALTHCHECK --interval=30s CMD wget -q -O-
http://localhost:8080/actuator/health || exit 1
ENTRYPOINT ["java", "-jar", "app.jar"]
```

4.5 Kubernetes Deployment & HPA

```
# k8s/deployment.yaml
apiVersion: apps/v1
kind: Deployment
metadata: { name: cyberq-app, namespace: production }
spec:
  replicas: 3
  strategy: { type: RollingUpdate, rollingUpdate: { maxSurge: 1,
maxUnavailable: 0 } }
  template:
    spec:
      securityContext: { runAsNonRoot: true, runAsUser: 1000 }
      containers:
      - name: cyberq-app
        image: registry.gitlab.com/seclogic/cyberq:latest
        resources:
          requests: { cpu: "250m", memory: "256Mi" }
          limits: { cpu: "500m", memory: "512Mi" }
        livenessProbe: { httpGet: { path: /actuator/health/liveness, port:
8080 }, initialDelaySeconds: 60 }
        readinessProbe: { httpGet: { path: /actuator/health/readiness, port:
8080 }, initialDelaySeconds: 30 }
```

Namespace	Workloads	CPU / Memory Quota	Purpose
production	cyberq-app (3 replicas)	4 vCPU / 4Gi	Production application workloads
staging	cyberq-app (1 replica)	2 vCPU / 2Gi	Pre-production testing
monitoring	prometheus, grafana, alertmanager	3 vCPU / 6Gi	Observability stack
logging	elasticsearch, logstash, kibana, filebeat	4 vCPU / 8Gi	Log management stack

Table 4.2: Kubernetes Namespace and Resource Allocation

4.6 Monitoring – Prometheus & Grafana

```
# prometheus/alerting-rules.yaml
groups:
- name: cyberq-alerts
  rules:
  - alert: PodCrashLooping
    expr: rate(kube_pod_container_status_restarts_total[15m]) > 0
    for: 5m
    labels: { severity: critical }
    annotations: { summary: "Pod {{ $labels.pod }} is crash-looping" }
  - alert: HighCPUUsage
    expr: container_cpu_usage_seconds_total > 0.8
    for: 5m
    labels: { severity: warning }
    annotations: { summary: "CPU > 80% for 5 minutes" }
```

[Fig. 4.3: Grafana Dashboard – System Metrics Overview – Insert Screenshot Here]

Fig. 4.3: Grafana Dashboard – System Metrics Overview

4.7 Logging – ELK Stack

```
# logstash/pipeline/main.conf
input { beats { port => 5044 } }
filter {
  if [kubernetes][namespace] == "production" {
    grok { match => { "message" => "%{TIMESTAMP_ISO8601:ts} %{LOGLEVEL:level}
%{GREEDYDATA:msg}" } }
    date { match => ["ts", "ISO8601"]; target => "@timestamp" }
  }
}
output { elasticsearch { hosts => ["elasticsearch:9200"]; index => "cyberq-logs-%{+YYYY.MM.dd}" } }
```

[Fig. 4.4: Kibana Log Analysis Dashboard – Insert Screenshot Here]

Fig. 4.4: Kibana Log Analysis Dashboard

4.8 Security Scanning – Trivy

```
# Trivy image scan
trivy image \
  --severity CRITICAL,HIGH \
  --exit-code 1 \
  --ignore-unfixed \
  --format json \
  --output trivy-report.json \
  registry.gitlab.com/seclogic/cyberq:${IMAGE_TAG}
```

[Fig. 4.5: Trivy Security Scan Output – Insert Screenshot Here]

Fig. 4.5: Trivy Security Scanning – Vulnerability Report

Chapter 5

Testing and Results

5.1 Testing Methodology

A comprehensive testing strategy was adopted covering: unit testing (pipeline stage and manifest validation), integration testing (end-to-end pipeline execution), security testing (Trivy scan accuracy and RBAC enforcement), performance testing (HPA response under load via JMeter), and deployment testing (Kubernetes rollout correctness and health probe compliance). All results were tracked against acceptance criteria defined in Chapter 2.

5.2 Test Cases and Results

5.2.1 CI/CD Pipeline Test Cases

TC ID	Test Description	Expected	Actual Result	Status
TC-01	Pipeline triggers on git commit to main	Triggered < 30s	Triggered in ~12s	PASS
TC-02	Maven build – all unit tests pass	BUILD SUCCESS	47/47 tests passed	PASS
TC-03	Pipeline abort on build failure	Fail + alert sent	FAILED, Slack alert sent	PASS
TC-04	Feature branch – no deploy stage	Deploy stage skipped	Stage skipped correctly	PASS
TC-05	End-to-end pipeline execution time	< 15 minutes	11 min 43 sec	PASS

Table 5.1: Test Cases – CI/CD Pipeline Execution

5.2.2 Docker & Security Scanning Test Cases

TC ID	Test Description	Expected	Actual Result	Status
TC-06	Multi-stage Docker build	Image < 200MB	178MB	PASS
TC-07	Trivy scan – clean production image	0 CRITICAL/HIGH	0 Critical, 0 High	PASS
TC-08	Trivy scan – image with known CVEs	Pipeline FAILS	exit-code 1, FAILED	PASS
TC-09	Container runs as non-root user	UID != 0	UID = 1000 confirmed	PASS
TC-10	Image push to GitLab registry	Push successful	Image available in registry	PASS

Table 5.2: Test Cases – Docker Build and Security Scanning

5.2.3 Kubernetes Deployment Test Cases

TC ID	Test Description	Expected	Actual Result	Status
TC-11	Rolling update – zero downtime	No requests dropped	No downtime observed	PASS
TC-12	All pods reach READY 1/1 state	All pods READY	3/3 pods READY	PASS
TC-13	HPA scales under CPU load	Scale to ≥ 8 replicas	Scaled to 8 in ~90s	PASS
TC-14	Rollback on failed deployment	Rollback completes	Rollback in 45s	PASS
TC-15	RBAC rejects unauthorised API call	HTTP 403	403 Forbidden returned	PASS

Table 5.3: Test Cases – Kubernetes Deployment

5.2.4 Monitoring & Logging Test Cases

TC ID	Test Description	Expected	Actual Result	Status
TC-16	Prometheus scrapes /metrics endpoints	All targets UP	All targets UP	PASS
TC-17	Grafana renders all dashboard panels	All panels populated	14/14 panels rendered	PASS
TC-18	Alert fires on CPU threshold breach	Alert + notification sent	Alert triggered & sent	PASS
TC-19	Kibana query returns structured log fields	All fields extracted	All fields verified	PASS
TC-20	Logstash grok parsing accuracy	Fields parsed correctly	Parsing verified	PASS

Table 5.4: Test Cases – Monitoring and Logging

5.3 Overall Test Execution Summary

Test Category	Total	Passed	Failed	Pass Rate
CI/CD Pipeline Testing	5	5	0	100%
Docker & Security Scanning	5	5	0	100%
Kubernetes Deployment	5	5	0	100%
Monitoring & Alerting	3	3	0	100%
ELK Stack Logging	2	2	0	100%
TOTAL	20	20	0	100%

Table 5.5: Overall Test Execution Summary

All twenty test cases achieved a PASS result (100% pass rate). The CI/CD pipeline consistently executed in 12–14 minutes. Trivy successfully blocked all intentionally vulnerable test images. The Kubernetes HPA scaled from 3 to 8 replicas within 90 seconds under load, and the ELK Stack provided reliable structured log extraction across all validation scenarios.

[Fig. 5.1: Successful CI/CD Pipeline Execution Output – Insert Screenshot Here]

Fig. 5.1: Successful CI/CD Pipeline Execution Output

Chapter 6

Conclusion

The internship project "CyberQ Shield (CNAPP Features)" at Seclogic Technology Private Limited has been a technically comprehensive engagement that bridged academic knowledge with production-grade implementation across DevOps, cloud security, and cloud-native engineering.

All seven defined objectives were successfully achieved. The fully automated CI/CD pipeline — validated through twenty test cases at 100% pass rate — demonstrated consistent execution within 12–14 minutes. The Trivy scanning gate operationalised "shift-left" security, blocking insecure images in all test scenarios. Terraform-based IaC established reproducible, auditable AWS infrastructure management. The Kubernetes deployment configuration with HPA demonstrated responsive scaling from 3 to 8 replicas under load. The Prometheus/Grafana and ELK Stack observability infrastructure enabled real-time anomaly detection and in-depth log analysis.

From a personal perspective, this internship significantly strengthened competencies in Linux administration, YAML configuration management, cloud networking, Kubernetes operations, and security engineering. The CyberQ Shield project demonstrates that integrating security, automation, and observability within a unified CNAPP framework is not only technically feasible but operationally transformative — providing a replicable DevSecOps blueprint for cloud-native environments.

Chapter 7

Future Scope

7.1 Full GitOps with ArgoCD

A complete GitOps implementation with ArgoCD would declare all Kubernetes workload states in Git, with the operator continuously reconciling actual cluster state — providing automated drift detection, correction, and intuitive Git-based rollback across environments.

7.2 AI-Driven Anomaly Detection

Integrating machine learning via Elastic ML or Prometheus Anomaly Detector would enable detection of subtle metric and log deviations indicative of security intrusions or failures, beyond static threshold-based alerting.

7.3 Advanced DevSecOps – SAST, DAST & SCA

Incorporating SonarQube (SAST), OWASP ZAP (DAST), and Dependency-Track (SCA) would provide comprehensive supply chain and application security coverage beyond container image scanning.

7.4 Service Mesh with Istio

Istio would provide mutual TLS for all inter-service communication, fine-grained traffic management, circuit breaking, and distributed tracing — enhancing security and observability of microservices within the cluster.

7.5 Multi-Cloud Deployment & Policy as Code

Extending Terraform modules to support multi-cloud deployments (AWS, Azure, GCP) and implementing OPA/Gatekeeper for Policy as Code would eliminate vendor lock-in and automate compliance enforcement at the Kubernetes API level.

7.6 eBPF-Based Runtime Security with Falco

Falco or Tetragon (eBPF-powered) would provide real-time detection of anomalous container system call behaviour — detecting in-progress attacks, privilege escalation, and container escapes beyond what static pipeline scanning provides.

References

- [1] Kim, G., Humble, J., Debois, P., & Willis, J. (2016). The DevOps Handbook. IT Revolution Press.
- [2] Burns, B. et al. (2016). Borg, Omega, and Kubernetes. ACM Queue, 14(1), 70-93.
- [3] Humble, J., & Farley, D. (2010). Continuous Delivery. Addison-Wesley Professional.
- [4] Poulton, N. (2023). Docker Deep Dive. Independently Published.
- [5] Gartner Research. (2021). Innovation Insight for Cloud-Native Application Protection Platforms.
- [6] HashiCorp. (2024). Terraform Documentation. <https://developer.hashicorp.com/terraform/docs>
- [7] Kubernetes Community. (2024). Kubernetes Documentation. <https://kubernetes.io/docs/home/>
- [8] Jenkins Community. (2024). Jenkins Pipeline Syntax. <https://www.jenkins.io/doc/book/pipeline/>
- [9] GitLab Inc. (2024). GitLab CI/CD Documentation. <https://docs.gitlab.com/ee/ci/>
- [10] Aqua Security. (2024). Trivy Scanner Documentation. <https://trivy.dev/>
- [11] Prometheus Authors. (2024). Prometheus Documentation. <https://prometheus.io/docs/>
- [12] Grafana Labs. (2024). Grafana Documentation. <https://grafana.com/docs/grafana/latest/>
- [13] Elastic N.V. (2024). ELK Stack Documentation. <https://www.elastic.co/guide/>
- [14] Amazon Web Services. (2024). AWS Well-Architected Framework. <https://aws.amazon.com/architecture/well-architected/>
- [15] NIST. (2022). SP 800-190: Application Container Security Guide.
- [16] CIS. (2023). CIS Kubernetes Benchmark v1.8.0. Center for Internet Security.
- [17] Beyer, B. et al. (2016). Site Reliability Engineering. O'Reilly Media.
- [18] Argo Project. (2024). ArgoCD Documentation. <https://argo-cd.readthedocs.io/>
- [19] Open Policy Agent. (2024). OPA Documentation. <https://www.openpolicyagent.org/docs/>
- [20] Weaveworks. (2022). Guide to GitOps. <https://www.weave.works/technologies/gitops/>

Appendix A – Abbreviations and Acronyms

Abbreviation	Full Form
AWS	Amazon Web Services
CI/CD	Continuous Integration / Continuous Delivery
CNAPP	Cloud-Native Application Protection Platform
CSPM	Cloud Security Posture Management
CVE	Common Vulnerabilities and Exposures
CWPP	Cloud Workload Protection Platform
ELK	Elasticsearch, Logstash, Kibana
HPA	Horizontal Pod Autoscaler
IaC	Infrastructure as Code
KSPM	Kubernetes Security Posture Management
RBAC	Role-Based Access Control
SAST	Static Application Security Testing
VPC	Virtual Private Cloud
YAML	YAML Ain't Markup Language

Appendix B – Sample Configuration Files

B.1 Terraform Variables

```
# variables.tf
variable "aws_region" { default = "us-east-1" }
variable "ubuntu_ami_id" { default = "ami-0c7217cdde317cfec" }
variable "key_pair_name" { description = "EC2 Key Pair name" }
variable "environment" { default = "production" }
```

B.2 Kubernetes Service

```
# k8s/service.yaml
apiVersion: v1
kind: Service
metadata: { name: cyberq-app-svc, namespace: production }
spec:
  type: LoadBalancer
  selector: { app: cyberq-shield }
  ports: [{ port: 80, targetPort: 8080 }]
```

B.3 HPA Configuration

```
# k8s/hpa.yaml
apiVersion: autoscaling/v2
kind: HorizontalPodAutoscaler
metadata: { name: cyberq-app-hpa, namespace: production }
spec:
  scaleTargetRef: { apiVersion: apps/v1, kind: Deployment, name: cyberq-app }
  minReplicas: 3
  maxReplicas: 10
  metrics:
  - type: Resource
    resource: { name: cpu, target: { type: Utilization, averageUtilization: 70 } }
```